Curriculum Vitae — Nico Lang

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Research experience

03/2025–	**Assistant Professor** — Computer Vision and Machine Learning for Earth Observation, *Department of Computer Science (DIKU), University of Copenhagen; Global Wetland Center; Pioneer Centre for AI*
09/2022–03/2025	**Postdoctoral Researcher**, *DIKU, University of Copenhagen; Pioneer Centre for AI*
02/2018–05/2022	**PhD Research Assistant**, *Photogrammetry and Remote Sensing, ETH Zürich*
03/2013–10/2017	**Student Researcher**, *Photogrammetry and Remote Sensing, ETH Zürich*
02/2015–04/2015	**Intern**, *LOGITECH UPICTO GmbH*
10/2014–01/2015	**Intern**, *upicto GmbH (ETH spin-off)*

Academic degrees

05/2022	Doctor of Sciences, *ETH Zürich*
09/2017	MSc in Geomatics, *ETH Zürich*
08/2014	BSc in Geomatik und Planung, *ETH Zürich*

Scientific focus areas

Computer vision; Deep learning; Machine learning; Remote sensing; Earth observation; Environmental monitoring; Global vegetation structure mapping; Representation learning; Self-supervised learning, Open-world recognition; Uncertainty quantification; Fine-grained recognition

Selected professional activities

2026–	**ELLIS member**
2025–	**Area Chair** for CVPR 2026 and ECCV 2026
2025–	**Program Committee** (Keynotes) *Nordic AI Meet 2025*
2025–	**Co-director** *P1 program on AI for Climate & Conservation*
2025–	**Coordinator** *Climate & Conservation at the Pioneer Centre for AI*
2025–	**Core team member** *Climate AI Nordics network*
2024	**Organizer** *PhD summer course SSL4EO-2024: Self-Supervised Learning for Earth Observation, University of Copenhagen*
2023	Drummer in the *CVPR 2023 house band*, Vancouver
	Workshop organization
2026	**Organizer** *Climate AI Nordics workshop, Copenhagen*
2026	**Organizer** *AICC-2: Workshop on AI for Climate and Conservation, ECCV*
2025	**Organizer** *AICC: Workshop on AI for Climate and Conservation, EurIPS*
2025	**Organizer** *Advances in Representation Learning for Earth Observation (REO), EurIPS*
2024	**Organizer** *Visipedia workshop 2024*, Copenhagen
2023–	**Organizer** (main point of contact) *FGVC10, FGVC11, FGVC12, FGVC13, CVPR*

Academic services

2019–	**Reviewer** *CVPR* [**Outstanding reviewer 2023**], *ECCV, ICCV, TPAMI, RSE, ISPRS, Nature Comms. Env., Global Change Biology, NLDL, CVPR FGVC workshop, NeurIPS CompSust-2023 workshop, ICCV Joint workshop on marine vision, AI in Science Summit 2025, EurIPS AICC workshop, EurIPS REO workshop*
2027	**External examiner** for Nils Lehmann's PhD thesis, *Technical University of Munich* - [upcoming]
2026	**External examiner** for Miguel Espinosa's PhD thesis, *University of Edinburgh*
2026	**External examiner** for Yuchang Jiang's PhD thesis, *University of Zurich*
2025	**Reviewer** PhD proposal by Claire Robin, *Wageningen University & Research*
2025	**Reviewer** for the ELLIS PhD Program

Bibliography overview

Published 22 peer-reviewed journal and conference articles. See Google Scholar or the separate publication list. Note that CVPR, ECCV, NeurIPS, ICLR are top-tier conferences ranked A* according to CORE.

Ten selected publications

[1] Lucia Gordon, Serge Belongie, Christian Igel, and **Lang, Nico**. "MMEarth-Bench: Global Model Adaptation via Multimodal Test-Time Training". In: *ECCV*. 2026.

[2] Sander Riisøen Jyhne, Christian Igel, Morten Goodwin, Per-Arne Andersen, Serge Belongie, and **Lang, Nico**. "SuperF: Neural Implicit Fields for Multi-Image Super-Resolution". In: *ICLR*. 2026.

[3] Yuyan Chen, **Lang, Nico**, B. Christian Schmidt, Aditya Jain, Yves Basset, Sara Beery, Maxim Larrivée, and David Rolnick. "Open-Insect: Benchmarking Open-Set Recognition of Novel Species in Biodiversity Monitoring". In: *NeurIPS Datasets and Benchmarks Track*. [**Spotlight**]. 2025.

[4] Vishal Nedungadi, Ankit Kariryaa, Stefan Oehmcke, Serge Belongie, Christian Igel, and **Lang, Nico**. "MMEarth: Exploring multi-modal pretext tasks for geospatial representation learning". In: *ECCV*. 2024.

[5] **Nico Lang**, Vésteinn Snæbjarnarson, Elijah Cole, Oisin Mac Aodha, Christian Igel, and Serge Belongie. "From coarse to fine-grained open-set recognition". In: *CVPR*. 2024.

[6] Bingchen Zhao, **Lang, Nico**, Serge Belongie, and Oisin Mac Aodha. "Labeled Data Selection for Category Discovery". In: *ECCV*. 2024.

[7] Nikolai Kalischek, **Lang, Nico**, Cécile Renier, Rodrigo Caye Daudt, Thomas Addoah, William Thompson, Wilma J Blaser-Hart, Rachael Garrett, Konrad Schindler, and Jan D Wegner. "Cocoa plantations are associated with deforestation in Côte d'Ivoire and Ghana". In: *Nature Food* (2023).

[8] Siyu Liu, Martin Brandt, Thomas Nord-Larsen, Jerome Chave, Florian Reiner, **Lang, Nico**, Xiaoye Tong, Philippe Ciais, Christian Igel, Adrian Pascual, et al. "The overlooked contribution of trees outside forests to tree cover and woody biomass across Europe". In: *Science Advances* (2023).

[9] **Nico Lang**, Walter Jetz, Konrad Schindler, and Jan Dirk Wegner. "A high-resolution canopy height model of the Earth". In: *Nature Ecology & Evolution* (2023).

[10] **Nico Lang**, Nikolai Kalischek, John Armston, Konrad Schindler, Ralph Dubayah, and Jan Dirk Wegner. "Global canopy height regression and uncertainty estimation from GEDI LIDAR waveforms with deep ensembles". In: *Remote sensing of environment* (2022).

Talks

Optimizing representations at test time - From super-resolution to semantic features

2026 Keynote *Open-Earth-Monitor Global Workshop 2026*, Barcelona

2026 Invited talk *CVPR 2026 workshop on Monitoring the World Through an Imperfect Lens*, Denver

Learning Representations for Global Ecosystem Modelling

2027 Keynote *CVPR 2027 EarthVision workshop*, Seattle - *[upcoming]*

2027 Invited talk *ICLR 2027 workshop on Machine Learning for Remote Sensing*, San Francisco - *[upcoming]*

2026 Keynote *BMVC Workshop on Machine Vision for Earth Observation, Environmental Monitoring, and Climate Change* , Lancaster - *[upcoming]*

2026 Invited talk *ECCV 2026 workshop on Computer Vision for Ecology*, Malmö

2026 Keynote *ELLIS Winter School 2026: AI for Earth System, Hazards & Climate Extremes*, Athens

Fusing Remote Sensing Data Across Sensors and Time

2025 Seminar talk *Wageningen University & Research*, Wageningen

Learning From Global Earth Observation Data

2025 Keynote *OpenGeoHub summer school "Data Science for Earth observation"*, Wageningen

2025 Keynote *QIM Workshop on "Challenges in analysis of large image data", SCIA2025*, Reykjavík

2024 Invited talk *Winter school "BigAI for BigGIS", Indian Institute of Remote Sensing*, Webinar

2024 Invited talk *Workshop LifeCLEF2024, Grenoble*

2024 Seminar talk *Allen Institute for AI (AI2)*, Seattle

2024 Seminar talk *Danish Hydraulic Institute (DHI)*, Hørsholm

MMEarth: Multi-Modal Pretext Tasks For Geospatial Representation Learning

2025 Invited talk *"From Pixels to Products: CV in Danish Research & Industry", D3A*, Nyborg
2025 Invited talk *Nordic Workshop on AI for Climate Change*, Gothenburg
2025 Talk *ESA-NASA International Workshop on AI Foundation Model for EO*, Frascati

Global Vegetation Monitoring With Probabilistic Deep Learning
2023 Invited talk *RISE Learning Machines Seminars*, Lund
2023 Invited talk *AI for Good*, Webinar
2023 Keynote *PhD course on "Deep learning in forest remote sensing"*, Ås
2022 Invited talk *ML4RS: Machine learning for remote sensing reading group*, Webinar

Forest Canopy Height Mapping at Global Scale by Fusing Sentinel-2 and Gedi
2022 Talk *ESA Living planet symposium*, Bonn

Global Vegetation Height Mapping With Deep Ensembles
2022 Seminar talk *Alan Turing Institute*, Webinar

High Carbon Stock Mapping at Large Scale
2022 Lightning talk *Google Geo for Good*, Webinar

Towards Global High-Resolution Biomass Maps: Machine Vision at Large Scale
2021 Seminar talk *ESA Φ-Lab*, Webinar
2021 Invited talk *High carbon stock approach (HCSA) Largescale Working Group*, Webinar

Machine Learning for Environmental Sciences
2019 Invited talk *WSIS Forum Workshop "Analysis of aerial data for Aid & Development"*, Geneva

Panel discussions

2026 *Career panel at the ELLIS Winter School 2026*, Athens
2025 *Nordic–Alan Turing Institute AI Research Workshop*, Oslo
2025 *OpenGeoHub Summer School "GeoAI – Dos and Don'ts"*, Wageningen
2025 *D3A Conference Session "From Pixels to Products"*, Nyborg
2024 *ICLR 2024 workshop ML for Remote Sensing, "Beyond Benchmarks: ML for the Planet"*, Vienna

Awards

2026 PAMI Mark Everingham Prize (FGVC team's contributions to the computer vision community)
2023 Culmann Prize (Outstanding doctoral thesis, ETH Zürich)
2023 CVPR Outstanding Reviewer
2019 U.V. Helava Award (Best Paper, ISPRS Journal)

Selected news media

For a complete list please refer to my personal website.

2022 German national TV ARD quiz show: Wer Weiss Denn Sowas
2022 Swiss national TV SRF news "10 vor 10": Living Planet Symposium mit Schweizer Beteiligung
2022 NASA: Scientists Show How Forests Measure Up
2022 NVIDIA: Neural Network Generates Global Tree Height Map, Reveals Carbon Stock Potential
2022 ETH Zürich: Neural network can read tree heights from satellite images
2021 SRF1 Swiss national radio live interview: Die Vermessung der Wälder
2021 Mongabay: Chocolate giant funds high resolution carbon map to protect forests
2020 ETH Zürich Industry Relations: A global tool against deforestation

Funding awarded (Total: ≈$355K)

2026 *Horizon Europe* - EarthGenerator a foundation model for Earth system modelling, (with Christian Igel), 282K Euro (Total project 6M Euro)
2026 *Voxel51* - Sponsorship for the FGVC13 workshop at CVPR (with Lukas Picek), $5000
2025 *Danish Data Science Academy* - Small event grant for the AICC EurIPS workshop, $7600.
2025 *Pioneer Centre for AI* - P1 program on AI for Climate & Conservation, $7700
2025 *Google DeepMind* - Sponsorship for the FGVC12 workshop at CVPR (with Serge Belongie and

Christine Kaeser-Chen), $5000

2025 *Pioneer Centre for AI* - Research assistant 4 months (part-time), $10,500 (in-kind)
Project: Neural Implicit Fields for Multi-Image Super-Resolution

2024 *Pioneer Centre for AI* - Part-time research assistant 2 months, $1,250 (in-kind)
Project: MMEarth-Bench: Global Environmental Tasks for Multimodal Geospatial Models

2024 *University of Copenhagen* - SSL4EO-2024 PhD summer course (with Stefan Oehmcke, Ankit Kariryaa, Christian Igel), $6800

2023 *Pioneer Centre for AI* - Research assistant 6 Months, $39,500 (in-kind)
Project: MMEarth: Exploring Multi-Modal Pretext Tasks For Geospatial Representation Learning

2023 *University of Copenhagen and University of Edinburgh (UoE)* - Strategic Partnership seed-fund (with Serge Belongie, Oisin Mac Aodha, Stella Frank), $17,700
Financed the Visipedia workshop 2024 and my 3-weeks research visit at UoE in April 2024.

2021 *Amazon* - Cloud computing credits (with Peter Gehler, Jan Dirk Wegner) $20,000
Project: A high-resolution canopy height model of the Earth

2018 *ETH Global* - Travel grant to represent ETH at the Clinton Global Initiative University meeting in Chicago, (with Andrés C Rodríguez), $2,000

2016 *Degen Stiftung ETH Zürich* - Travel award for summer research visit at Caltech (USA), $1,150

Supervisor functions

Co-supervised 12 master's and bachelor's theses. Co-advised 8 PhD students.

Teaching experience

09/2025 **Lecturer**, *OpenGeoHub Summer School "Data Science for EO"*, Wageningen

07/2024 **Lecturer**, *SSL4EO-2024 PhD Summer Course, University of Copenhagen*

02/2018–05/2022 **Teaching Assistant**, *Photogrammetry and Remote Sensing, ETH Zürich*

03/2013–09/2017 **Teaching Assistant**, *ETH Zürich*

08/2010–08/2017 **Volleyball Instructor**, *J+S School Sport*, Switzerland

Tutorials on "Deep Learning for Geospatial Data Analysis"

09/2019 Tutorial, *ECML/PKDD Summer School (EPSS19)*, Würzburg

06/2019 Lecture & Tutorial, *ISPRS Geospatial Week*, Enschede

01/2019 Tutorial, *Workshop on ML for Environmental and Geosciences (MLEG2019)*, Zürich

01/2018 Tutorial, *ISPRS TC II Symposium, Riva*

Selected mentorships

2024- **PhD student (University of Agder & The Norwegian Mapping Authority)** Co-supervising Sander Riisøen Jyhne since his one-year research visit at the University of Copenhagen. Together we are working on implicit neural representations for multi-image super-resolution. Publication at ICLR 2026.

2024- **PhD student (Harvard & University of Copenhagen)** Co-supervising Lucia Gordon (with Andrew Davies, Harvard) since her summer visit at the Pioneer Centre for AI in 2024. As a follow-up of the MMEarth project, we are working on the MMEarth-Bench project. Publication at ECCV 2026.

2024- **PhD student (McGill University & Mila Quebec AI Institute)** Co-advising Yuyan Chen (with David Rolnick, McGill & Sara Beery, MIT) working on benchmarking open-set recognition and category discovery for insect monitoring. Publication at NeurIPS 2025, Spotlight.

2023- **PhD student (University of Copenhagen)** Co-supervising Hui Zhang at together with Christian Igel. Hui is working on an extension of my PhD research modelling vertical vegetation structure globally. Publication under review.

2024- **Research Assistant (Pioneer Centre for AI)** Supervised Vishal Nedungadi as a research assistant after his master, working on the MMEarth project for 6 months at the Pioneer Centre for AI, which was published at ECCV 2024. I mentored Vishal during the application for PhD positions. Co-advising Vishal in his PhD research at Wageningen University & Research.

2023-2025 **SURF undergrad (Caltech)** Supervised Avirath Sundaresan during a summer research visit

as a SURF fellow. Mentored Avirath during weekly meetings together with Sara Beery (MIT) working on animal re-identification.

2023-2024 **Bachelor students (University of Copenhagen)** Co-supervised Philip Enevoldsen and Christian Gundersen during their Bachelor thesis. Mentored them beyond their thesis to present at the ICCV 2023 workshop OOD-CV and the NLDL 2025 conference leading to a publication in PMLR proceedings, supported by a DDSA travel grant.

Programming & Software

Python: PyTorch, Rasterio, GeoPandas
Other bash, Slurm, GDAL, QGIS

Languages

English (fluent), German (native), Swiss-german

References

Serge Belongie
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Professor of Computer Science
Director, Pioneer Centre for AI
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Christian Igel
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Konrad Schindler
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Jan Dirk Wegner
University of Zürich
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Sara Beery
Massachusetts Institute of Technology
Homer A. Burnell Career Development Professor
Co-lead, Global Center on AI and Biodiversity Change
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Oisin Mac Aodha
University of Edinburgh
Reader (Associate Professor) in Machine Learning
oisin.macaodha@ed.ac.uk

Ralph Dubayah
University of Maryland
Professor of Geographical Sciences
NASA GEDI Principal Investigator
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Last updated: October 30, 2026